

DEPARTMENT OF COMPUTER SCIENCE • SINGLE SUBJECT SERIES

CS-302

COMPUTER NETWORKS MASTER TEXTBOOK

Comprehensive 23-Page University Lecture Notes, Theory Foundations &
Solved Exam Question Bank

OFFICIAL 23-PAGE SINGLE SUBJECT EDITION • VERIFIED ACADEMIC MASTERPIECE

CURRICULAR SCOPE & CORE MODULES INCLUDED:

- OSI 7-Layer Reference Model vs TCP/IP Protocol Stack
- Application Layer: HTTP/HTTPS, DNS, SMTP, FTP, Sockets
- Transport Layer: TCP 3-Way Handshake, Flow Control, UDP
- Network Layer: IPv4/IPv6, CIDR Subnetting, Routing (Distance Vector & Link State)
- Data Link Layer: Framing, Error Detection (CRC, Parity), CSMA/CD

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SYLLABUS**CS-302: Computer Networks**

Protocol architectures, packet switching, transport layer flow control, IP addressing, subnetting, and network security.

COURSE OVERVIEW

Layered network architectures (OSI 7-Layer and TCP/IP models), data link protocols, IP addressing (IPv4/IPv6, CIDR subnetting), routing protocols (OSPF, BGP), transport protocols (TCP flow and congestion control vs UDP), application layer (DNS, HTTP/1.1, HTTP/2, TLS), and network security.

1.0 Accredited Learning Outcomes**Outcome 1**

Deconstruct packet transmission across the physical, link, network, transport, and application layers.

Outcome 2

Calculate CIDR subnet masks, usable host ranges, and broadcast addresses for complex enterprise networks.

Outcome 3

Trace TCP 3-way handshakes, sequence acknowledgment arithmetic, and congestion control states.

Outcome 4

Inspect application protocols (DNS lookups, HTTP headers, TLS handshakes) using packet capture analyzers.

NAVIGATION

Comprehensive Table of Contents

Detailed Index across all 23 Pages of CS-302

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CHAPTER 1**Network Layering Models, Protocol Architectures & Encapsulation**

OSI 7-Layer Reference Model vs TCP/IP Internet Suite

Computer networks rely on layered abstraction: each layer provides services to the layer above while shielding implementation details below. During transmission, data moves down the stack, with each layer prepending a protocol header (encapsulation); at the destination, headers are stripped sequentially (decapsulation).

Theoretical Foundation 1.1

OSI 7-Layers: Physical, Data Link, Network, Transport, Session, Presentation, Application.

Theoretical Foundation 1.2

TCP/IP 4-Layers: Network Interface (Link), Internet (IP), Transport (TCP/UDP), Application (HTTP, DNS).

Theoretical Foundation 1.3

PDU (Protocol Data Unit) Terminology: Frame (Link), Packet/Datagram (Network), Segment (Transport), Message/Data (Application).

SYSTEM ARCHITECTURE

Detailed architectural trace.

OSI Layer	TCP/IP Equivalent	Key Protocols	Addressing Scheme
Application (7)	Application	HTTP, DNS, SMTP, SSH	URLs, Hostnames, Mailboxes
Presentation (6)	Application	TLS/SSL, JPEG, ASCII	Data Formats & Encodings
Session (5)	Application	RPC, NetBIOS, Sockets	Session Identifiers
Transport (4)	Transport	TCP, UDP, QUIC	Port Numbers (0-65535)
Network (3)	Internet	IPv4, IPv6, ICMP, BGP	IP Addresses (32-bit / 128-bit)
Data Link (2)	Network Access	Ethernet 802.3, Wi-Fi 802.11	MAC Addresses (48-bit hex)
Physical (1)	Network Access	Fiber Optic, Cat6e, 5G Radio	Bits, Volts, Frequencies

SAMPLE EXAM QUESTION: Midterm Calculation

A network is assigned IP block 192.168.10.0/24. Subnet this network into 4 equal subnets. State the subnet mask, network address, usable host range, and broadcast address for each.

SOLUTION & KEY TAKEAWAY

To create 4 subnets, we borrow 2 bits from host portion ($2^2 = 4$). New mask is /26 (255.255.255.192). Block size is $256 - 192 = 64$.

Subnet 1: Net 192.168.10.0/26, Usable: .1 to .62, Broadcast: .63

Subnet 2: Net 192.168.10.64/26, Usable: .65 to .126, Broadcast: .127

Subnet 3: Net 192.168.10.128/26, Usable: .129 to .190, Broadcast: .191

Subnet 4: Net 192.168.10.192/26, Usable: .193 to .254, Broadcast: .255.

CHAPTER 2**Transport Layer: TCP Reliability, Flow & Congestion Control**

Three-Way Handshake, Sliding Window, Slow Start & Fast Retransmit

IP provides unreliable, best-effort packet delivery. TCP constructs a reliable, in-order byte stream over this unreliable foundation using checksums, sequence numbers, positive acknowledgments, sliding window flow control, and dynamic congestion control algorithms.

Theoretical Foundation 2.1

TCP 3-Way Handshake: SYN -> SYN-ACK -> ACK establishes sequence sync and receiver window parameters.

Theoretical Foundation 2.2

TCP Connection Teardown: FIN -> ACK -> FIN -> ACK with TIME_WAIT state ($2 \times \text{MSL}$) to drain duplicate segments.

Theoretical Foundation 2.3

Flow Control vs Congestion Control: Flow control prevents overflowing the receiver buffer (advertised window); Congestion control prevents overflowing internet routers (congestion window cwnd).

Theoretical Foundation 2.4

Congestion Algorithms: Slow Start (exponential growth), Congestion Avoidance (additive increase), Fast Retransmit (3 duplicate ACKs), Fast Recovery.

SYSTEM ARCHITECTURE

Detailed architectural trace.

SAMPLE EXAM QUESTION: Final Exam Mastery

Why does the TCP connection teardown process include a TIME_WAIT state of 2MSL (Maximum Segment Lifetime)?

SOLUTION & KEY TAKEAWAY

- 1) To ensure the final ACK sent by the client arrives safely at the server; if the final ACK is lost, the server will retransmit its FIN, which the client can only re-acknowledge if it remains in TIME_WAIT.
- 2) To allow all old duplicate segments from the connection to dissipate completely in the network, preventing them from being misinterpreted by a future connection utilizing the same port pair.

LECTURE NOTES

CS-302 Slide Deck: Core Modules 1 & 2

Classroom Lecture Companion

SLIDE 1: MODULE 1: NETWORK ARCHITECTURES & THE OSI MODEL

- Packet switching vs Circuit switching: statistical multiplexing vs dedicated reservation
- OSI 7 layers: Physical, Data Link, Network, Transport, Session, Presentation, Application
- TCP/IP 4-layer model: Link, Internet (IP), Transport (TCP/UDP), Application
- Encapsulation / Decapsulation: Headers and PDUs (Frame, Packet, Segment, Message)

SLIDE 2: MODULE 2: APPLICATION LAYER & SOCKET PROTOCOLS

- HTTP/1.1 persistent connections vs HTTP/2 multiplexing vs HTTP/3 (QUIC over UDP)
- DNS resolution hierarchy: Root, TLD (.com), Authoritative DNS servers (recursive vs iterative)
- Socket API basics: `socket()`, `bind()`, `listen()`, `accept()`, `connect()`, `send()`, `recv()`
- Email protocols: SMTP (push to mail server) vs IMAP/POP3 (pull from mail server)

SLIDE 3: MODULE 3: TRANSPORT LAYER: TCP VS UDP

- TCP: Reliable, connection-oriented, byte-stream, in-order delivery; UDP: Unreliable, datagram
- TCP 3-Way Handshake: SYN -> SYN-ACK -> ACK; 4-Way Teardown: FIN -> ACK -> FIN -> ACK
- TCP Flow Control: Sliding window advertised by receiver (rwnd)
- TCP Congestion Control: Slow Start, Congestion Avoidance, Fast Retransmit, Fast Recovery

SLIDE 4: MODULE 4: IPV4, CIDR & SUBNET CALCULATIONS

- IPv4 addresses: 32 bits (4 octets); Classful addressing (Class A, B, C) vs CIDR notation (/24)
- Subnet Mask calculation: Network bits (1s) and Host bits (0s)
- Number of usable hosts = $2^{(\text{host_bits})} - 2$ (subtract Network ID and Broadcast ID)
- Network address = IP & SubnetMask; Broadcast address = IP | (~SubnetMask)

KEY EXAM HIGHLIGHTS FROM LECTURES

1. Master CIDR Subnetting calculations (network address, broadcast address, valid host ranges) under time pressure.
2. Trace TCP sequence and acknowledgment numbers during connection establishment and data transfer.
3. Contrast Distance Vector (Bellman-Ford, count-to-infinity) with Link State (Dijkstra) routing.

LAB & PRACTICUM CHECKLIST

Project Option 1: Multi-client Chat Server using TCP Sockets in C++ or Python

Project Option 2: Packet Sniffer & HTTP Header Analyzer using Raw Sockets

Project Option 3: Subnet Calculator and IP Range Visualizer

CHAPTER 6**CS-302 Midterm Examination Solved Bank**

Step-by-step Model Solutions

MIDTERM PROBLEM [Final Exam Mastery]

Compare Distance Vector Routing (Bellman-Ford) and Link State Routing (Dijkstra) in terms of convergence and routing loops.

OFFICIAL MODEL SOLUTION

Distance vector routers share their entire routing table only with immediate neighbors; they suffer from slow convergence and the Count-to-Infinity problem. Link state routers flood local link state advertisements (LSAs) to the entire network so every node builds an identical topology map, converging rapidly without routing loops.

CHAPTER 6 (CONT.)**CS-302 Final Examination Mastery**

Comprehensive Rubrics

EXAMINATION PROTOCOL & SCORING RUBRICS

1. Allocate time proportionally to marks.
2. Write formal preconditions and loop invariants.
3. Show scratch work for memory diagrams.
4. Clearly state time and auxiliary space complexity.

CHAPTER 7

CS-302 Semester Capstone Lab Specification

Hands-on Production Implementation

GRADING CRITERIA & DEMO EXPECTATIONS

- Correctness & Completeness: 40%
- Code Architecture & Clean Principles: 25%
- Memory Safety & Optimization: 20%
- Technical Documentation & Video Demo: 15%

CHAPTER 8**CS-302 Technical Interview Preparation**

FAANG & High-Growth Tech Interview Questions

INTERVIEW SCENARIO: Standard Senior Systems & Network Engineering Interview

What happens in precise network protocol sequence when you type `https://google.com` into your browser address bar and press Enter?

OPTIMAL ARCHITECTURAL SOLUTION

1. DNS resolution (browser cache -> OS cache -> recursive resolver -> authoritative root/TLD DNS via UDP 53).
2. ARP resolution to find local gateway default router MAC address.
3. TCP 3-way handshake on port 443 with server.
4. TLS 1.3 cryptographic handshake (ClientHello, ServerHello, certificate verification, ECDHE key exchange, session keys).
5. Encrypted HTTP/2 or HTTP/3 GET request sent over TLS.
6. Web server processes request, returns HTTP 200 OK HTML payload.
7. Browser renders DOM and fetches secondary assets.

INTERVIEW CHECKLIST

1. Clarify constraints and expected volume.
2. Explain algorithmic intuition before typing.
3. Identify space-time trade-offs.
4. Walk through an edge case manually.

CHAPTER 9

CS-302 Lexicon & Authoritative References

Glossary of Formal Terms and Literature

Key Term	Rigorous Academic Definition
MTU (Maximum Transmission Unit)	The largest packet size that can be transmitted over a network medium without fragmentation (typically 1500 bytes on Ethernet).
BGP (Border Gateway Protocol)	The exterior gateway protocol that routes traffic across autonomous systems (AS) on the global Internet backbone.

- RECOMMENDED TEXTBOOKS
1. Computer Networking: A Top-Down Approach - Kurose & Ross

2. Computer Networks - Andrew S. Tanenbaum & David J. Wetherall

3. TCP/IP Illustrated - W. Richard Stevens